KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 6 — IMPLEMENTATION STRATEGY

How can KrishiGrid be realistically deployed across Bangladesh?

A strong system is not just designed it must be built step by step.

Analysis : Pilot districts, rollout phases, hub deployment logic
Tools : SQL, Power BI
Datasets : pilot_districts.csv + lands.csv + hub_costs.csv

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THE IMPLEMENTATION STRATEGY

THE QUESTION

How do we move from concept to real-world implementation?

APPROACH

- Start with high-potential pilot districts across different area types
- Use phased rollout instead of nationwide deployment at once
- Prioritize ward-level hubs before scaling upward
- Align infrastructure with local crop patterns and land types
- Integrate cost, impact, and feasibility into deployment decisions

WHAT WE MEASURE

- Pilot district feasibility and cost
- Number of hubs required (ward, union, district)
- Initial economic impact (oil, income, jobs)
- Payback period at pilot level
- Scalability across regions

DATASET OVERVIEW

File : pilot_districts.csv
Rows : Selected pilot districts
Columns : District profile + implementation metrics

Key Fields:

- *District name and division*
- *Area type (char, haor, coastal, plain, etc.)*
- *Estimated number of hubs (ward, union, district)*
- *Implementation cost per district*
- *Expected economic impact (income, oil, jobs)*

Source: *Derived from lands.csv + hub_costs + impact_projection*

SQL Queries

Question
Why are the selected pilot districts suitable for launching KrishiGrid?

```
SELECT
    district_name,
    division,
    area_type,
    total_agricultural_land_hectare,
    total_untapped_land_hectare,
    estimated_farmer_households,
    main_crops
FROM pilot_districts;
```

Key Insight:

The selected pilot districts represent a wide range of Bangladesh's agricultural conditions, including coastal (Patuakhali), plain (Rangpur), barind (Rajshahi), char-coastal (Noakhali), and hilly (Rangamati) regions, ensuring that the model is tested across diverse agro-ecological zones. These districts also show substantial agricultural capacity, with land ranging from ***85,000 to 210,000*** *hectares and untapped potential of* ***13,000 to 25,000*** *hectares, indicating strong opportunity for production expansion.*
*At the same time, each district supports a large farming population (****95,000 to 220,000*** *households) and diverse crop patterns, from rice and oilseeds to vegetables and spices. This combination of diversity, scale, and untapped potential makes these districts ideal for piloting KrishiGrid, as they allow validation of the system under real and varied conditions before national rollout.*

Question
How many hubs are required in the selected pilot districts to implement KrishiGrid?

```
SELECT
district_name,
ward_hubs_needed,
union_hubs_needed,
district_hubs_needed,
(ward_hubs_needed + union_hubs_needed +
district_hubs_needed) AS total_hubs
FROM pilot_districts;

-- Total hubs across all pilot districts
SELECT
SUM(ward_hubs_needed) AS total_ward_hubs,
SUM(union_hubs_needed) AS total_union_hubs,
SUM(district_hubs_needed) AS total_district_hubs,
SUM(ward_hubs_needed + union_hubs_needed +
district_hubs_needed) AS total_hubs
FROM pilot_districts;
```

Key Insight:

The pilot rollout requires a total of **2,275** hubs, of which **1,874** are ward-level hubs, **396** union hubs, and only 5 district hubs, clearly showing a strong bottom-up architecture. More than **80%** of the infrastructure is concentrated at the ward level, ensuring that services are physically close to farmers and aligned with the core design principle of accessibility.
At the district level, variation reflects scale and geography, with districts like Rangpur and Noakhali requiring over **530+** hubs, while smaller or hilly regions like Rangamati require fewer (**280** hubs), demonstrating adaptive deployment based on local conditions.
Overall, this structure confirms that KrishiGrid is designed as a decentralized and scalable system, where the majority of investment directly impacts farmers rather than being concentrated in higher-level infrastructure.

SQL Queries

Question
What is the total investment required to implement the KrishiGrid pilot across selected districts?

```
-- District-wise cost
SELECT
district_name,
total_ward_hub_cost_crore_bdt,
total_union_hub_cost_crore_bdt,
total_district_hub_cost_crore_bdt,
total_pilot_cost_crore_bdt
FROM pilot_districts;

-- Total pilot investment
SELECT
SUM(total_ward_hub_cost_crore_bdt) AS total_ward_cost,
SUM(total_union_hub_cost_crore_bdt) AS total_union_cost,
SUM(total_district_hub_cost_crore_bdt) AS
total_district_cost,
SUM(total_pilot_cost_crore_bdt) AS total_pilot_cost
FROM pilot_districts;
```

Key Insight:

The total investment required to launch the KrishiGrid pilot across 5 districts is approximately 1,200 crore BDT, covering diverse agro-regions and a large farming population. A significant portion of this cost around 1,022 crore BDT (≈85%) is allocated to ward-level hubs, confirming that most investment is directly focused on farmer-facing infrastructure rather than higher-level facilities.
At the district level, costs vary based on scale, with Rangpur requiring the highest investment (~331 crore BDT) due to larger coverage, while smaller regions like Rangamati require much less (~122 crore BDT), showing adaptive cost allocation. Overall, the pilot achieves wide geographic and economic coverage at a relatively manageable cost, making it a financially feasible starting point for national rollout.

Question
What economic impact will the KrishiGrid pilot generate in the first year across selected districts?

```
SELECT
   district_name,
   year1_additional_oil_production_mt,
   year1_farmer_income_increase_pct,
   year1_post_harvest_loss_reduction_pct,
   year1_direct_jobs_created,
   estimated_annual_hub_revenue_crore_bdt
FROM pilot_districts;

SELECT
   SUM(year1_additional_oil_production_mt) AS
total_oil_production_mt,
   AVG(year1_farmer_income_increase_pct) AS
avg_income_increase_pct,
   AVG(year1_post_harvest_loss_reduction_pct) AS
avg_loss_reduction_pct,
   SUM(year1_direct_jobs_created) AS total_jobs_created,
   SUM(estimated_annual_hub_revenue_crore_bdt) AS
total_annual_revenue
FROM pilot_districts;
```

Key Insight:

The KrishiGrid pilot across 5 districts delivers immediate and measurable impact in the first year, generating approximately 46,500 MT of additional oil production, creating over 10,800 direct jobs, and producing around 155 crore BDT in annual revenue. At the same time, farmers experience an average income increase of nearly 8%, while post-harvest losses reduce by about 3.5%, showing early efficiency gains even at pilot scale.
What makes this significant is not just the numbers, but the speed of impact within just one year, the system begins generating production, income, and employment simultaneously. This confirms that KrishiGrid is not a long-gestation project, but a fast-acting economic system capable of delivering real results from the initial rollout phase itself.

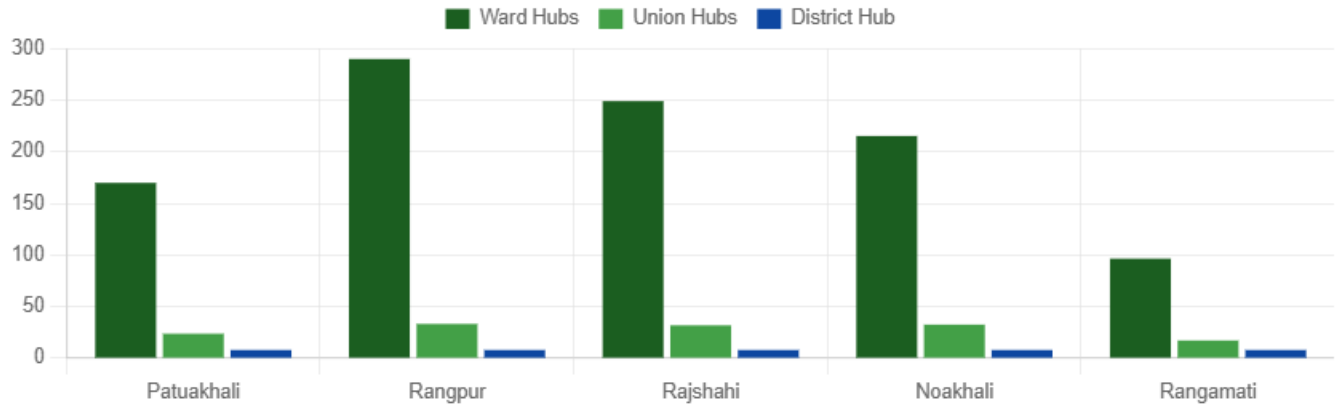
1200.8
Total Pilot Investment
Crore BDT

1,874
Total Ward Hubs (Pilot)
Hubs

10,814
Total Year 1 Jobs
Direct Jobs

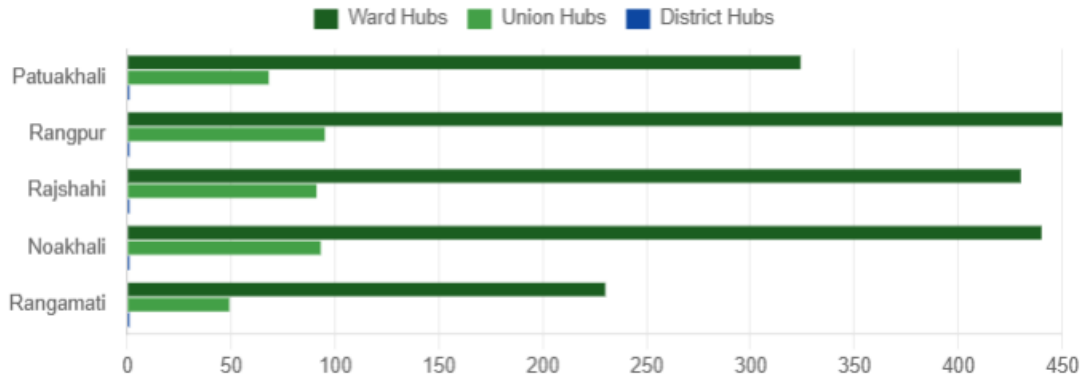
46,500
Total Year 1 Oil Production
MT

Pilot Investment Breakdown by District (Crore BDT)



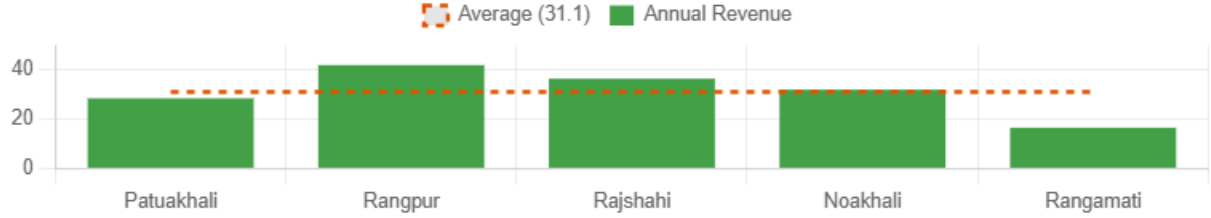
Over 85% of pilot investment is at ward level — directly farmer-facing. Rangpur requires the highest investment due to its larger geographic scale.

Hub Count by District



Pilot requires 1,874 ward hubs, 396 union hubs, and 5 district hubs. Ward hubs dominate confirming the bottom-up design philosophy of KrishiGrid.

Estimated Annual Hub Revenue by District (Crore BDT)



Patuakhali and Rangpur generate the highest annual hub revenue due to high agricultural output. All 5 districts show positive revenue from Year 1.

Investment Share by District

Rangpur accounts for the largest share of pilot investment due to scale. Rangamati the smallest due to the lower-cost hilly zone hub variant.

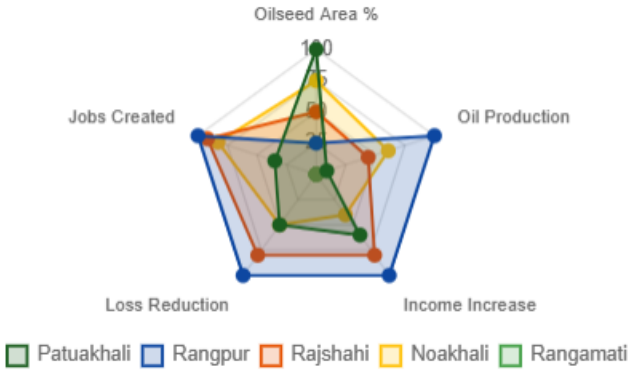
7.9
Avg Farmer Income Increase
% (Year 1)

3.6
Avg Post-Harvest Loss Reduction
% (Year 1)

Patuakhali
Fastest Payback District
7.1 years payback

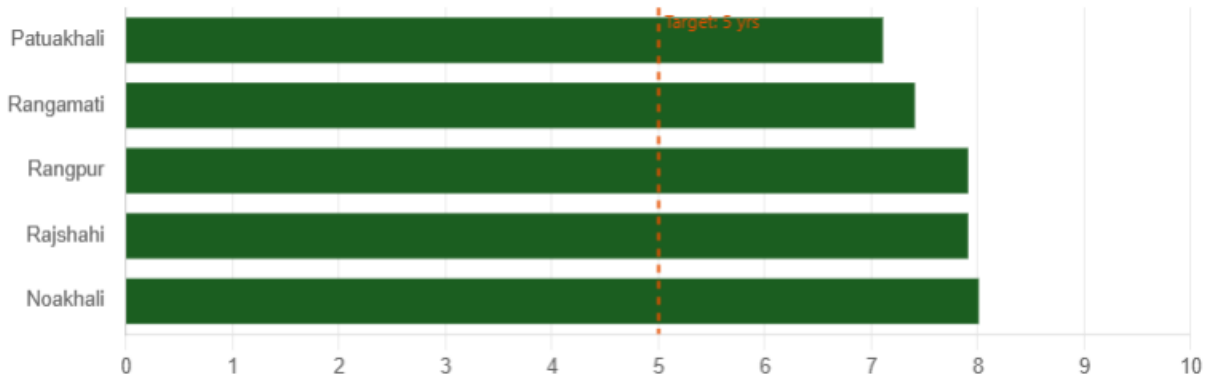
46,500
Total Year 1 Oil Production
MT across 5 districts

Year 1 Impact Comparison Across Districts



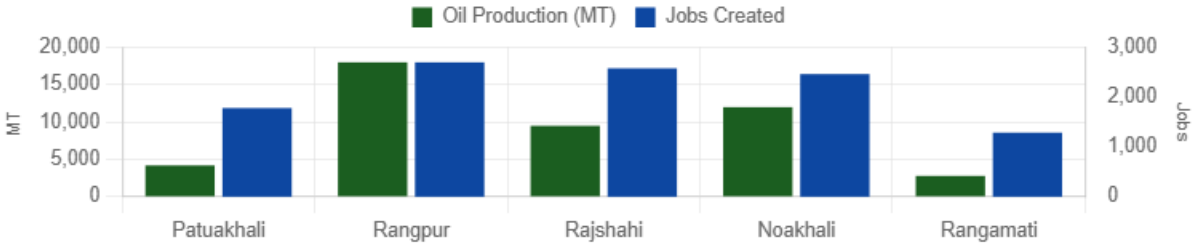
Each pilot district excels in different metrics. Patuakhali leads in oil production, Rangpur in jobs. Diversity of outcomes validates all hub variants.

Payback Period by District (Years)



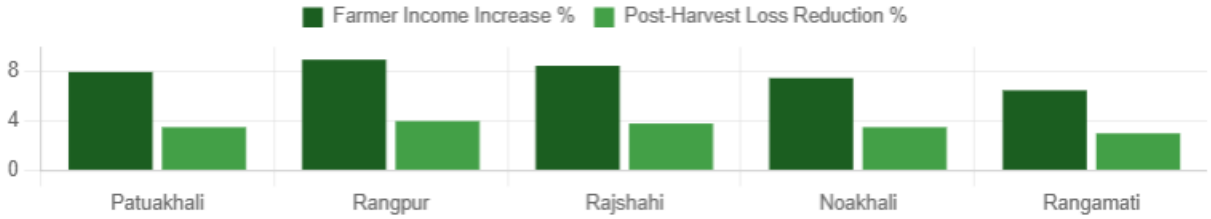
Patuakhali has the shortest payback due to high oilseed potential. Even Rangamati's longer payback fills a critical infrastructure gap with no current alternative.

Year 1 Oil Production and Job Creation by District



All 5 districts generate both additional oil and new jobs in Year 1, confirming KrishiGrid is productive from the very first year of operation.

Farmer Income Increase and Post-Harvest Loss Reduction (Year 1 %)



Farmer income increases and post-harvest loss reductions are consistent across all districts in Year 1. The hub model delivers reliable results regardless of region type.

KEY INSIGHTS

1. KrishiGrid succeeds not just because of infrastructure, but because of its system design — aligning production, processing, and market access into one coordinated network

2. The model solves structural inefficiencies rather than increasing production, unlocking hidden value already present in the system

3. A decentralized ward-first approach ensures maximum farmer participation (≈90%) compared to traditional centralized systems (~30%)

4. Government-backed pricing at hub level creates a strong price anchor, reducing exploitation by middlemen and stabilizing local markets

5. Interconnected hubs form a national food grid, allowing surplus regions to support deficit regions and preventing artificial shortages

6. The system combines public infrastructure with operational flexibility, making it suitable for government, PPP, or cooperative execution models

7. Real-time data flow across hubs enables better decision-making, demand forecasting, and supply coordination at national scale

8. The model is inherently scalable — starting from pilot districts and expanding gradually without disrupting existing systems

9. By integrating processing at source, the system ensures value addition stays with farmers, increasing rural income and economic resilience

10. The combination of physical flow (crops), reverse flow (inputs & price signals), and data flow creates a complete agricultural ecosystem

11. KrishiGrid complements existing megaprojects by activating economic productivity rather than replacing infrastructure

12. Overall, the system transforms agriculture from a fragmented sector into a coordinated economic network, ensuring long-term food security, farmer prosperity, and national stability